Introduction

Quiz Management & Online Assessment Platform

1. Introduction

The **Quiz Management & Online Assessment Platform** is a web-based application that allows students or users to participate in online quizzes and assessments.

The platform will have **two roles: Admin and Student/User**.

The **Admin** will have complete control over the platform, including creating quizzes, managing questions, managing users, monitoring quiz attempts, and viewing overall performance.

The **Student/User** will be able to browse available quizzes, attempt quizzes, view results, review answers, and track their performance.

The project is designed to provide practical experience with **frontend development, backend development, authentication, authorization, CRUD operations, database management, REST APIs, timers, scoring systems, dashboards, and responsive web design**.

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2. Project Objectives

The main objectives are:

- Build a complete online quiz platform.
- Implement secure user authentication.
- Implement Admin and Student roles.
- Allow Admin to create and manage quizzes.
- Allow Admin to create and manage questions.
- Allow students to attempt quizzes online.
- Implement automatic scoring.
- Implement a countdown timer.
- Store quiz attempts and results.
- Provide performance analytics.
- Provide a leaderboard.
- Create separate Admin and Student dashboards.
- Build a responsive web application.

User Roles

3. User Roles

The application will have only **two roles**.

3.1 Admin

The Admin has complete control over the platform.

Admin Features

- Admin login
- Admin dashboard
- Manage students/users
- Create quizzes
- Edit quizzes
- Delete quizzes
- Publish/unpublish quizzes
- Create questions
- Edit questions
- Delete questions
- Manage categories
- Manage difficulty levels
- View all quiz attempts
- View individual student results
- View quiz performance
- View platform analytics
- Manage leaderboard
- Activate/deactivate users

3.2 Student/User

Students can participate in quizzes available on the platform.

Student Features

- Register
- Login
- Logout
- View available quizzes

- Search quizzes
- Filter quizzes
- View quiz details
- Start quiz
- Answer questions
- Navigate between questions
- View remaining time
- Submit quiz
- View result
- Review answers
- View previous attempts
- Track performance
- View leaderboard

Main Application Modules

4. Main Application Modules

Module 1: Authentication

The application should provide authentication for Admin and Students.

Student Authentication

- Registration
- Login
- Logout
- Forgot password
- Reset password

Admin Authentication

- Admin login
- Logout
- Password reset

Admin accounts can either be created manually by the system owner or through a secure admin-management mechanism.

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5. Admin Dashboard

The Admin Dashboard will be the central control panel of the application.

Dashboard Statistics

The Admin should be able to see:

- Total students
- Total quizzes
- Published quizzes
- Draft quizzes
- Total questions
- Total quiz attempts
- Average score

- Total passed attempts
- Total failed attempts

Dashboard Analytics

Display charts for:

- Quiz attempts over time
- Student registrations
- Average quiz scores
- Pass/fail ratio
- Most popular quizzes
- Most popular categories

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6. User Management

The Admin can manage all registered students.

Features

- View all students
- Search students
- View student profile
- View student's quiz history
- View student's performance
- Activate/deactivate account
- Delete account

Student Information

Name
Email
Registration Date
Account Status
Quizzes Attempted
Average Score
Highest Score

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7. Quiz Management

All quiz management will be performed by the Admin.

Create Quiz

Admin can create a quiz containing:

- Quiz title
- Description
- Category
- Difficulty
- Duration
- Passing percentage
- Maximum attempts
- Quiz status
- Thumbnail/image (optional)

Example:

Quiz: JavaScript Fundamentals

Category: JavaScript
Difficulty: Intermediate
Duration: 20 minutes
Passing Score: 60%
Maximum Attempts: 2
Status: Published

Quiz Status

A quiz can have:

- Draft
- Published
- Unpublished

Only **published quizzes** should be available to students.

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8. Category Management

Admin can manage quiz categories.

Examples:

- HTML
- CSS
- JavaScript
- React
- Node.js
- Python
- Java
- Database
- Computer Networks
- Cyber Security

Admin Features

- Create category
- Edit category
- Delete category
- View quizzes under category

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9. Question Management

Admin can manage questions for every quiz.

Question Information

Each question can contain:

- Question text
- Options
- Correct answer
- Explanation
- Marks
- Difficulty

Example:

Question

Which method converts a JSON string into a JavaScript object?

A. JSON.stringify()

B. JSON.parse()

C. JSON.convert()

D. JSON.object()

Correct Answer: JSON.parse()

Explanation: JSON.parse() converts a JSON string into a JavaScript object.

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10. Question Types

The initial version can support:

Multiple Choice Question

One correct answer.

What does HTML stand for?

○ Hyper Text Markup Language
○ High Text Machine Language
○ Hyper Tool Multi Language
○ Home Tool Markup Language

Future Enhancement

The platform can later support:

- Multiple correct answers
- True/False
- Fill in the blanks
- Match the following
- Image-based questions
- Code-based questions

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11. Quiz Attempt System

Students can start any published quiz.

The quiz interface should display:

JavaScript Fundamentals

Time Remaining: 14:32

Question 5 of 20

Which keyword is used to declare a constant?

○ var
○ let
○ const
○ static

[Previous] [Next]

[Submit Quiz]

The student should be able to:

- Select an answer
- Move to the next question
- Move to the previous question
- Navigate directly to a question
- See answered/unanswered questions
- Submit the quiz

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12. Timer System

Every quiz can have a predefined duration.

Examples:

- 10 minutes
- 20 minutes
- 30 minutes
- 60 minutes

The timer should:

- Start when the quiz starts.
- Display remaining time.
- Continue correctly after page refresh where possible.
- Automatically submit the quiz when the time expires.

The actual quiz start time and expiry time should be validated by the backend rather than relying only on the browser timer.

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13. Quiz Submission

When the student submits the quiz, the backend should process the submission.

The system should calculate:

- Correct answers
- Incorrect answers
- Unanswered questions
- Total marks
- Obtained marks
- Percentage
- Pass/fail status
- Time taken

Example:

Total Questions: 20

Correct Answers: 16
Incorrect Answers: 3
Unanswered: 1

Score: 80%

Status: PASSED

The scoring calculation should happen on the **backend** so students cannot manipulate their score through frontend code.

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14. Result System

After completing a quiz, students should receive a detailed result.

Result Page

```
--------------------------------
      QUIZ RESULT
--------------------------------

JavaScript Fundamentals

Score: 80%

Correct:       16
Incorrect:      3
Unanswered:     1

Time Taken: 14:21

Status: PASSED
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```

Students should also be able to review:

- Question
- Selected answer
- Correct answer
- Explanation
- Whether the answer was correct

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15. Student Dashboard

The Student Dashboard should provide an overview of the student's quiz activity.

Statistics

- Total quizzes attempted
- Total quizzes passed
- Total quizzes failed
- Average score
- Highest score
- Total questions answered

Dashboard Sections

Student Dashboard

Quizzes Attempted 15
Average Score 78%
Highest Score 96%
Passed 12
Failed 3

Recent Attempts

JavaScript Quiz 86%
React Quiz 92%
Python Quiz 74%

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16. Quiz Discovery

Students should be able to find quizzes easily.

Search

Search by:

- Quiz title
- Category

Filters

Filter by:

- Category
- Difficulty
- Duration
- Recently added
- Popularity

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17. Quiz Details Page

Before starting a quiz, students should see:

JavaScript Fundamentals

Description:
Test your knowledge of JavaScript fundamentals.

Category: JavaScript
Difficulty: Intermediate
Questions: 30
Duration: 30 Minutes
Passing Score: 60%
Maximum Attempts: 2

[Start Quiz]

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18. Quiz Attempt History

Students should be able to view their previous attempts.

Example:

Quiz	Date	Score	Status
JavaScript	04 Aug	86%	Passed
React	02 Aug	72%	Passed
Python	30 Jul	48%	Failed

Clicking an attempt should open its detailed result.

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19. Leaderboard

The platform can provide a leaderboard to increase engagement.

Rank students based on:

- Highest score
- Average score
- Number of quizzes completed

Example:

Rank	Student	Average Score
1	Rahul	96%
2	Priya	93%
3	Amit	91%

The leaderboard can be:

- Overall
- Category-wise
- Monthly
- Weekly

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20. Database Design

The project can use PostgreSQL/MySQL.

Users

users
---
id
name
email
password
role
status
created_at

The `role` field will contain:

ADMIN
STUDENT

Categories

categories
---
id
name
description
created_at

Quizzes

quizzes

---
id
title
description
category_id
difficulty
duration
passing_score
max_attempts
status
created_at
updated_at

Questions

questions
---
id
quiz_id
question_text
marks
explanation
difficulty
created_at

Options

options
---
id
question_id
option_text
is_correct

Attempts

attempts
---
id
quiz_id

user_id
score
percentage
correct_answers
incorrect_answers
unanswered
time_taken
status
started_at
completed_at

Answers

answers
---
id
attempt_id
question_id
selected_option_id
is_correct

Recommended Tech Stack

21. Recommended Tech Stack

Frontend

Recommended

- React.js
- Tailwind CSS

Additional Libraries

- React Router
- Axios
- Recharts
- React Hook Form

Backend

The project can be implemented using:

MERN-style JavaScript Stack

- Node.js
- Express.js

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22. Database

Recommended:

PostgreSQL

Alternative:

- MySQL
- MongoDB

PostgreSQL is recommended because the application has clear relationships between:

Users
↓
Attempts
↓
Quizzes
↓
Questions
↓
Options

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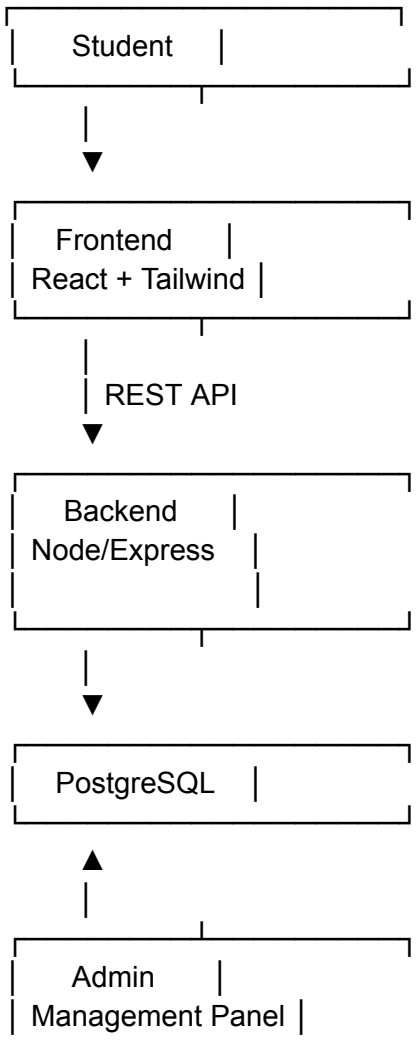
23. Application Architecture



Two-Week Development Schedule

24. Two-Week Development Schedule

Week 1

Day 1 – Project Setup

- Frontend setup
- Backend setup
- Database setup
- Git repository
- Environment configuration

Day 2 – Authentication

- Student registration
- Login
- Logout
- Password hashing
- JWT/session authentication

Day 3 – Role-Based Authorization

- Admin role
- Student role
- Protected routes
- Admin middleware
- Student middleware

Day 4 – Admin Dashboard

- Dashboard layout
- Statistics
- Navigation
- User management

Day 5 – Quiz Management

- Create quiz
- Edit quiz
- Delete quiz
- Publish/unpublish quiz

Day 6 – Category & Question Management

- Category CRUD
- Add questions
- Add options
- Correct answer
- Question editing/deletion

Day 7 – Student Quiz Interface

- Quiz listing
- Quiz details
- Start quiz
- Question navigation
- Answer selection
- Timer

Week 2

Day 8 – Quiz Submission

- Submit quiz
- Automatic submission
- Score calculation
- Pass/fail calculation

Day 9 – Results

- Result page
- Answer review
- Correct/incorrect answers
- Explanations
- Attempt history

Day 10 – Student Dashboard

- Statistics
- Quiz history
- Average score
- Performance charts

Day 11 – Admin Analytics

- Student statistics
- Quiz statistics
- Attempt statistics
- Pass/fail analytics

Day 12 – Leaderboard

- Ranking system
- Overall leaderboard
- Category leaderboard

Day 13 – Testing & Security

Test:

- Authentication
- Authorization
- Quiz creation
- Quiz attempts
- Timer
- Score calculation
- API validation
- Unauthorized access
- Input validation

Day 14 – Deployment & Documentation

- Deploy frontend
- Deploy backend
- Configure production database
- Environment variables
- Production testing
- README
- Screenshots
- Project presentation

Other info and Conclusion

25. Important API Endpoints

Authentication

POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
POST /api/auth/forgot-password
POST /api/auth/reset-password

Users – Admin Only

GET /api/users
GET /api/users/:id
PUT /api/users/:id
DELETE /api/users/:id
PATCH /api/users/:id/status

Categories – Admin Only

GET /api/categories
POST /api/categories
PUT /api/categories/:id
DELETE /api/categories/:id

Quizzes

GET /api/quizzes
GET /api/quizzes/:id

POST /api/quizzes
PUT /api/quizzes/:id
DELETE /api/quizzes/:id

PATCH /api/quizzes/:id/publish

`POST`, `PUT`, `DELETE`, and publishing operations should be restricted to Admin.

Questions – Admin Only

GET /api/quizzes/:quizId/questions
POST /api/quizzes/:quizId/questions
PUT /api/questions/:id
DELETE /api/questions/:id

Quiz Attempts – Student

POST /api/quizzes/:quizId/start
POST /api/quizzes/:quizId/submit
GET /api/attempts
GET /api/attempts/:id

Admin Results

GET /api/admin/attempts
GET /api/admin/attempts/:id
GET /api/admin/analytics

Leaderboard

GET /api/leaderboard

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26. Advanced Features

After completing the core requirements, students can implement additional features.

Question Randomization

Randomize questions for each attempt.

Option Randomization

Randomize answer options.

Negative Marking

Example:

Correct: +2
Wrong: -0.5
Skipped: 0

Maximum Attempts

Admin can configure the number of attempts allowed per quiz.

Quiz Scheduling

Admin can configure:

- Start date
- End date
- Availability

Certificate Generation

Generate a certificate when the student passes a quiz.

Email Notifications

Send emails for:

- Quiz completion
- Results
- Certificate generation

Dark Mode

Allow students and admins to switch between light and dark themes.

Question Import

Admin can upload questions using:

- CSV
- Excel

This can be a useful advanced feature for an internship-level project.

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27. Security Requirements

The application should implement:

- Password hashing
- JWT/session security
- Role-based authorization
- Input validation
- SQL injection prevention
- XSS prevention
- CSRF protection where applicable
- Rate limiting
- Secure HTTP headers
- API validation
- Environment variable protection
- Secure error handling

Important Rule

The frontend must **never be trusted** for:

- Correct answers
- Scores
- User roles
- Quiz completion status
- Attempt eligibility

These must be validated by the backend.

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28. Testing Requirements

Functional Testing

Test:

- Registration
- Login
- Admin login
- Quiz creation
- Question creation
- Quiz publishing
- Quiz attempt
- Quiz submission
- Score calculation
- Result generation

API Testing

Use:

- Postman
- Insomnia

Test:

- Valid requests
- Invalid requests
- Unauthorized requests
- Expired authentication
- Missing parameters
- Invalid IDs

Responsive Testing

The application should work properly on:

- Desktop

- Laptop

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29. Learning Outcomes

After completing this project, students will understand:

- Frontend development
- React.js
- Backend development
- REST APIs
- Database design
- PostgreSQL/MySQL
- Authentication
- Role-based authorization
- CRUD operations
- Form validation
- State management
- Timer implementation
- Scoring algorithms
- Dashboard development
- Data visualization
- API testing
- Web security
- Deployment
- Git/GitHub

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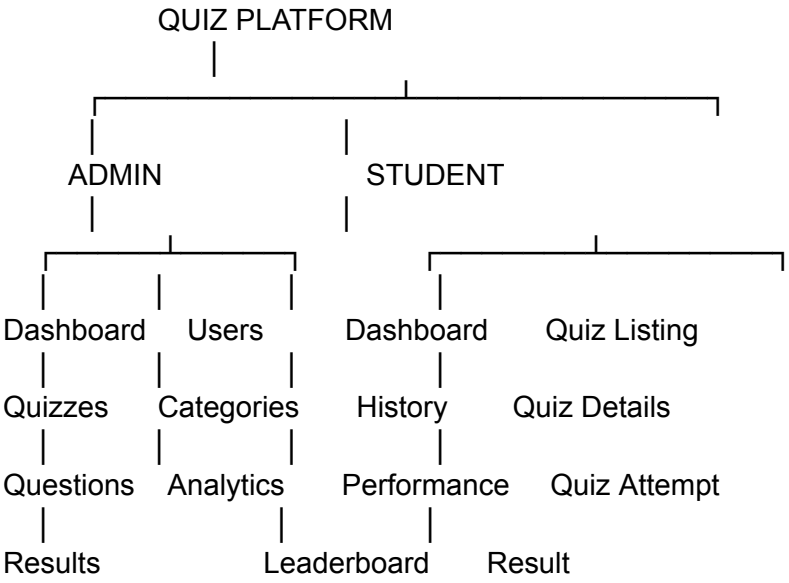
30. Expected Final Application

The completed application should contain:



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31. Conclusion

The **Quiz Management & Online Assessment Platform** is a complete full-stack web development project with a simple and practical role structure.

There are only two roles:

Admin → Manages the entire platform

Student → Takes quizzes and tracks performance

The Admin is responsible for creating quizzes, managing questions and categories, managing students, publishing quizzes, and monitoring results. Students can discover quizzes, attempt them within a time limit, receive automatic results, review their answers, and monitor their performance.

The project provides students with practical experience in **authentication, authorization, CRUD operations, REST APIs, relational databases, timers, scoring systems, dashboards, analytics, and web security**, making it suitable for an **internship-level or portfolio full-stack project**.